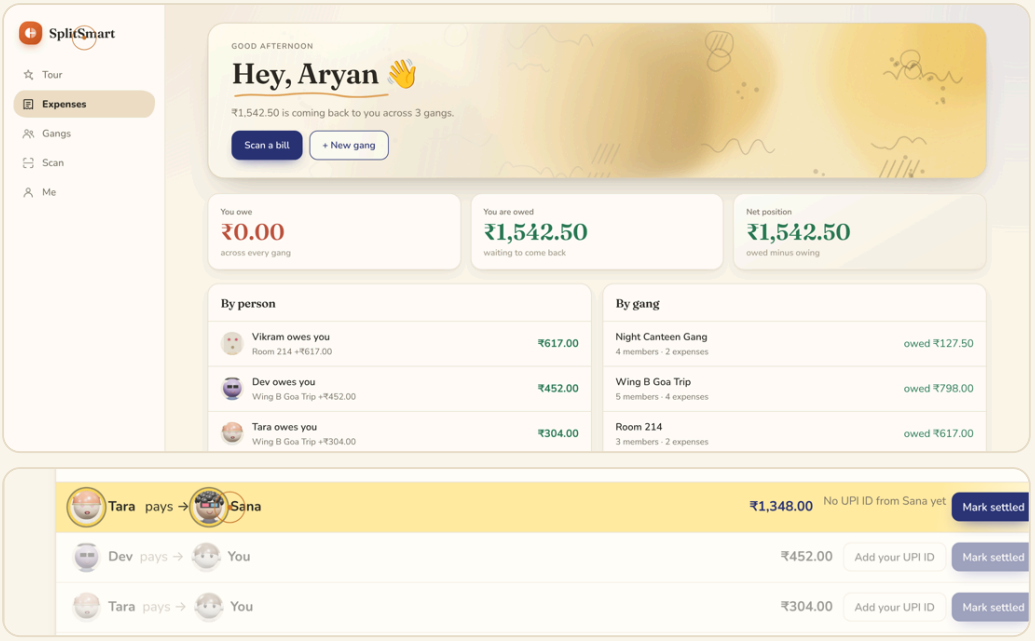
# SplitSmart

Scan a receipt, give each item to the people who actually shared it, and settle up with the fewest payments. One HTML file — no account, no backend, no build step, nothing leaves the device unless you share a gang.

Live
nothing4601.github.io/splitsmart

Code
github.com/Nothing4601/splitsmart

Tests
38 passing · node --test




Across every group, netted by person. Settle-up suggests the exact minimum set of payments; pointing at one shows who pays whom.

## Tier 1

CORE · 7 OF 7

- ✓ **Groups** — create, rename, delete; switch from one dashboard
- ✓ **Members** — add/remove, each with a 3D character avatar
- ✓ **Capture** — upload or camera, with preview before reading
- ✓ **OCR** — Tesseract in a worker; progress bar, UI stays live
- ✓ **Parsing** — name, qty, unit price, line total; subtotal, CGST/SGST, service, discount, total; addresses, GSTIN and footers dropped
- ✓ **Review** — every field editable, lines added/deleted, low-confidence words highlighted, photo beside the table
- ✓ **Validation** — warns with the exact difference when the lines don't reach the printed total
- ✓ **Manual expense** — description, amount, payer, participants, date
- ✓ **Balances** — each member's net, recomputed on every change

## Tier 2

ADVANCED · 7 OF 7

- ✓ **Itemized assignment** — tap avatars onto a line; shared items split between assignees; unassigned lines flagged before saving
- ✓ **Four split modes** — equal, exact, percent, shares; percentages must reach 100 and amounts must reach the bill
- ✓ **Proportional tax** — tax, service and discounts follow each person's item subtotal, not a head count
- ✓ **Rounding** — integer paise throughout; largest-remainder allocation, so shares always sum to the bill exactly
- ✓ **Debt simplification** — exact minimum payments by dynamic programming, not a greedy guess; mark any settled
- ✓ **Categories** — zero-shot model into the six categories, loaded on first use; your override always wins
- ✓ **Persistence** — groups, expenses and settlements in localStorage; receipt photos in IndexedDB

## Tier 3

BROWNIE POINTS · 9 OF 9

- ✓ **UPI settle-up** — upi://pay deep link and QR per payment, pre-filled
- ✓ **Analytics** — donut by category, bars by member, by group and by month
- ✓ **Share, no backend** — export/import JSON, or encode a group into a link
- ✓ **PWA & offline** — installable; every byte it needs ships with the site
- ✓ **Recurring** — monthly items add themselves each cycle
- ✓ **Export** — settlement summary as PDF, expenses as CSV
- ✓ **Batch scan** — a queue of receipts read in one go
- ✓ **Accessibility** — keyboard reachable, ARIA roles and labels, visible focus rings, honours reduced motion
- ✓ **Live deployment** — GitHub Pages, public URL

## Beyond the brief

Voice entry. Speak an expense — “Dev paid 600 for pizza, split equally” — or say who had what on a scanned bill: “I had the vada, the rest between the others.” Whisper runs on the device.

Live sync. Share a gang by link and two phones stay in step, AES-GCM encrypted through a public relay. No account, no server of ours. Join-by-link adds you as a member.

Cross-group net position. Everything you owe and are owed, netted across all groups and matched by person.

3D character avatars with a mix-and-match creator, a scroll-stacked feature tour, and a pointer layer that lights up both people in a settlement row.

## How it is built

**One file.** All of the app is `index.html` — CSS, markup and script inline. No bundler, no framework, no backend, nothing to install. three.js is vendored so neither the first paint nor offline use depends on a CDN.

**Money is never a float.** Everything is integer paise from parsing to settle-up. A documented largest-remainder rule hands out the odd paisa, so ₹100 three ways is 33.34 + 33.33 + 33.33 and the parts always rebuild the whole.

**The maths is tested, not trusted.** A DOM-free `split-logic` block holds allocation, itemised shares, balances, settle-up and the parsers. **38 unit tests** pull it out of the HTML and run it in a sandbox, covering rounding, tax proportioning, minimum-payment settlement and the natural-language cases.

**Nothing heavy on first paint.** OCR, the category model and Whisper are fetched only when first used.